

Name

Class



Year 9 Stage 5.3 Mathematics Yearly Examination 2023

Instructions

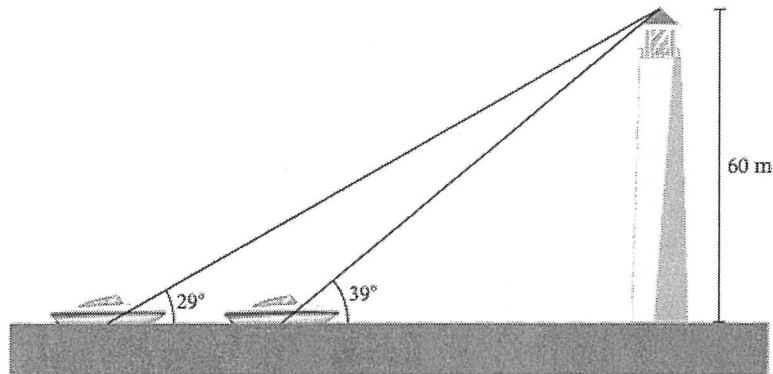
- Working time: 50 minutes
- Write in blue or black pen. You may draw diagrams in pencil.
- Show all working where necessary. Marks may be deducted for poor or illegible writing.
- NESA Approved Calculators are permitted for use.

All students MUST attempt all FOUR sections.

Trigonometry	Probability	Investigating Data	Simultaneous Equation	Total
/10	/10	/10	/12	/42

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1. There are TWO boats behind each other at sea. The angle of elevation from the **first** boat to the top of a lighthouse, 60 metres tall, is 29° . In addition, the angle of elevation from the **second** boat is 39° , as shown in the diagram.



- (a) Calculate the distance between the base of the lighthouse and the **first** boat, correct to 1 decimal place. 2

Let the distance be x .

Then

$$\tan 29^\circ = \frac{60}{x} \quad (1)$$

$$\therefore x = \frac{60}{\tan 29^\circ} \\ = \boxed{108.2 \text{ m}} \quad (1)$$

- (b) Similarly, calculate the distance between the base of the lighthouse and the **second** boat, correct to 1 decimal place. 1

Let the distance be y .

Then

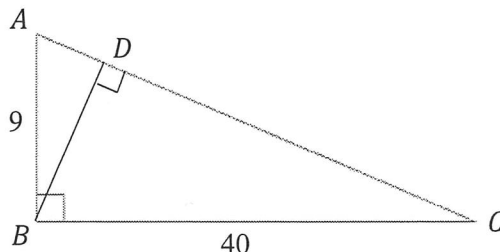
$$y = \frac{60}{\tan 39^\circ} \\ = \boxed{74.1 \text{ m}} \quad (1)$$

- (c) Hence, calculate the distance between the two boats, correct to 1 decimal place. 1

$$\text{distance} = x - y \\ = 108.2 - 74.1 \\ = \boxed{34.1 \text{ m}} \quad (1)$$

Section 1 continues on the following page.

2. ABC is a right-angled triangle where $\angle ABC = 90^\circ$ and $BD \perp AC$. The lengths of AB and BC are 9 and 40 respectively.



- (a) Calculate $\angle C$ to the nearest minute.

1

$$\tan C = \frac{9}{40}$$

$$\therefore C = \tan^{-1}\left(\frac{9}{40}\right) = 12^\circ 41' \quad (1)$$

- (b) Hence, or otherwise, calculate the length of BD , correct to 1 decimal place.

2

In $\triangle DBC$,

$$\sin C = \frac{BD}{40} \quad (1)$$

$$\therefore BD = 40 \sin 12^\circ 41'$$

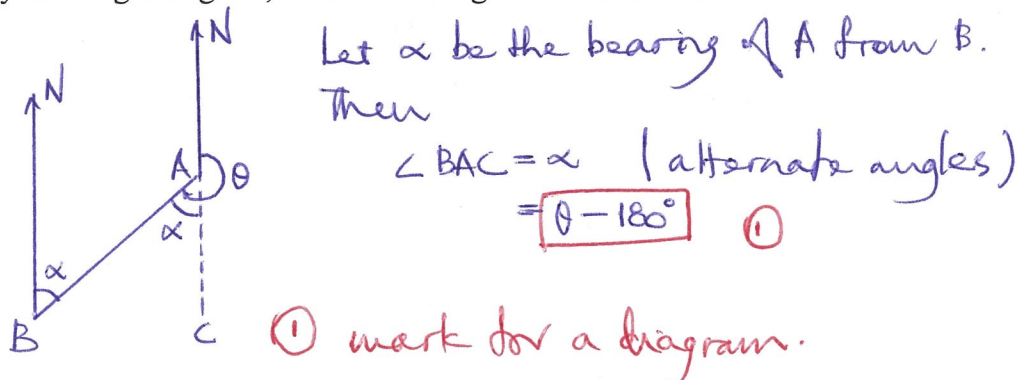
$$= 8.8 \quad (1)$$

3. Let the bearing of B from A be a reflex angle θ .

(Hint: A reflex angle is the angle between 180° and 360° .)

- (a) By drawing a diagram, find the bearing of A from B in terms of θ .

2



- (d) Hence, find the difference between the two bearings.

1

$$\text{difference} = \theta - \alpha$$

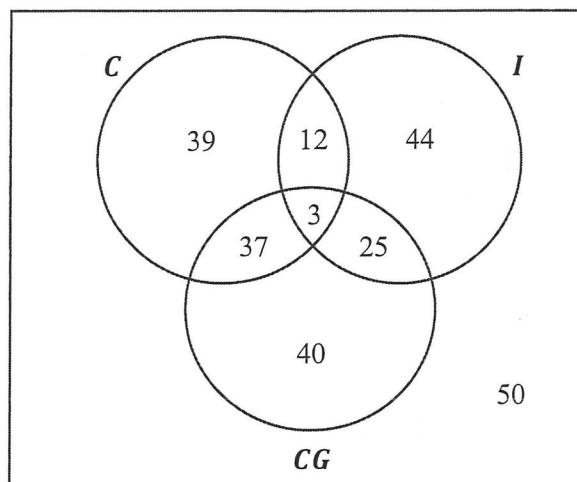
$$= \theta - (\theta - 180^\circ)$$

$$= 180^\circ \quad (1)$$

End of Section 1

1. A high school offers Chinese (*C*), Classical Greek (*CG*) and Italian (*I*) as language subjects for year 12 students for their HSC.

The last year's subject selection was shown in the following Venn diagram.



- (a) How many year 12 students were studying at the high school last year? 1

$$39 + 12 + 44 + 37 + 3 + 25 + 40 + 50 = \boxed{250} \quad (1)$$

- (b) Calculate the probability of selecting a student who studied at least one language subject. 1

$$\text{prob.} = 1 - \frac{50}{250} = \boxed{\frac{4}{5}} \quad (1)$$

- (c) Calculate the probability of selecting a student who studied Chinese **or** Classical Greek. 1

$$\text{prob.} = \frac{250 - 50 - 44}{250} = \boxed{\frac{78}{125}} \quad (1)$$

- (d) Calculate the probability of selecting a student who studied Classical Greek **and** Italian. 1

$$\text{prob.} = \frac{3 + 25}{250} = \boxed{\frac{14}{125}} \quad (1)$$

- (e) Given that a student studied at least two language subjects, calculate the probability that the student studied Chinese. 1

$$\text{prob.} = \frac{37 + 3 + 12}{37 + 3 + 25 + 12} = \boxed{\frac{52}{77}} \quad (1)$$

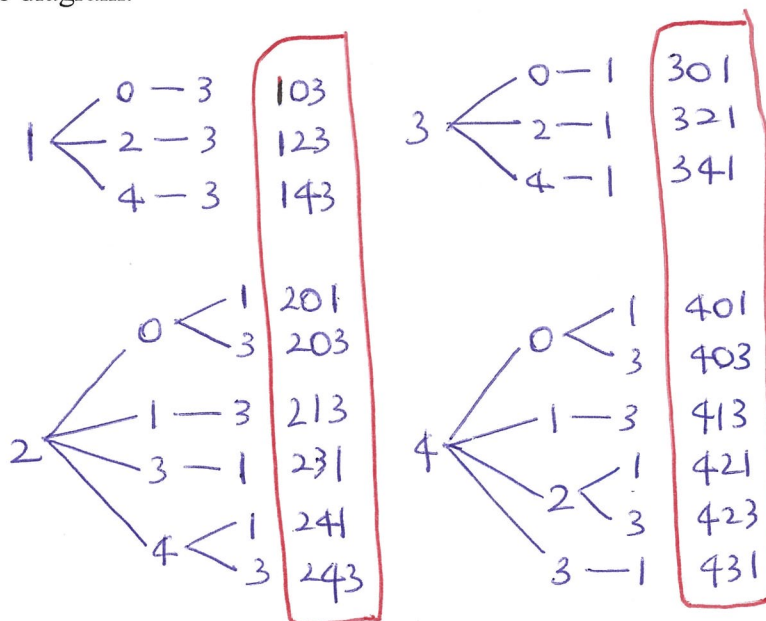
Section 2 continues on the following page.

2. From a set of cards numbered 0 to 4, three cards are drawn at random to form a three-digit **odd** number.

(Hint: Examples of odd numbers are 1, 3, 5, 7, 9, 11, 13, ...)

- (a) Show that there are 18 possible outcomes for the above event, using a tree diagram.

2



② marks for all correct outcomes or

① mark for at least 9 correct outcomes

- (b) Calculate the probability of having a prime number.

1

(Hint: A prime number is the number that is divisible by 1 and itself only.)

prime numbers are 103, 241, 401, 421, 431

$$\therefore \text{prob.} = \frac{5}{18} \quad \textcircled{1}$$

- (c) Calculate the probability of having a number whose digits are in descending order.

1

$$\text{prob.} = \frac{3}{18} = \frac{1}{6} \quad \textcircled{1} \quad (321, 421, 431)$$

- (d) Calculate the probability of having a **prime** number whose digits are in descending order.

1

$$\text{prob.} = \frac{2}{18} = \frac{1}{9} \quad \textcircled{1} \quad (421, 431)$$

End of Section 2

1. A movie received feedback scores from the audience using the scale of 1 (not interesting at all) to 5 (very interesting). The scores were as follows.

2	3	4	5	4
4	5	3	2	1
3	4	2	4	5

- (a) Fill in **all** the blanks in the following frequency table.

2

score (x)	frequency (f)	fx
1	1	1
2	3	6
3	3	9
4	5	20
5	3	15
total	15	51

② marks for all correct no.s.
or
① mark for all correct no.s in frequency column

- (b) Using the table in (a), find the mode and the mean of the scores.

2

$$\text{mode} = 4 \quad \text{①}$$

$$\text{mean} = \frac{51}{15} = \frac{17}{5} \quad \text{①}$$

- (c) Which of the following options best describes the types of the above data?

1

① categorical, discrete

② categorical, continuous

✓ ③ numerical, discrete

④ numerical, continuous

Section 3 continues on the following page.

2. Two examination (half-yearly and yearly) marks of 20 randomly chosen students were compared using a back-to-back stem-and-leaf plot.

Half-yearly		Yearly
9 6 3 2 0	5	5 8
8 6 5 3 4 2	6	3 4 9
7 5 4 2	7	0 2 3 6 8
① mark 8 3 0	8	2 4 5 7 7
1 0	9	1 5 9
	10	0 0

- (a) Calculate the mean of the **half-yearly** examination marks.

1

$$\text{mean} = \frac{1388}{20} = \boxed{69.4} \quad \textcircled{1}$$

- (b) Calculate the median of the **yearly** examination marks.

1

$$\text{median} = \frac{78 + 82}{2} = \boxed{80} \quad \textcircled{1}$$

- (c) Describe the shape of each examination marks.

2

Half-yearly exam $\Rightarrow$ positively skewed ①
 Yearly exam $\Rightarrow$ symmetrical ①

- (d) There is a mistake in the above plot. Circle it on the above plot and briefly explain why.

1

should be in descending order.

End of Section 3

1. (a) Solve the following simultaneous equations, using the **elimination** method.

3

$$3x - 4y = 41 \quad (1)$$

$$2x + 3y = 152 \quad (2)$$

$$(1) \times 2 - (2) \times 3$$

$$6x - 8y = 82$$

$$6x + 9y = 456 \quad (-)$$

$$-17y = -374$$

$$\therefore y = 22 \quad (1)$$

Substitute this into (1)

$$3x - 4 \times 22 = 41$$

$$\therefore 3x = 129$$

$$\therefore x = 43 \quad (1)$$

- (b) Solve the following simultaneous equations, using the **substitution** method.

3

$$2y - 3x = 8 \quad (1)$$

$$x + 3y = 1 \quad (2)$$

From (2),

$$x = 1 - 3y$$

Substitute this into (1)

$$2y - 3(1 - 3y) = 8$$

$$2y - 3 + 9y = 8$$

$$11y = 11$$

$$y = 1 \quad (1)$$

$$\therefore x = 1 - 3 \times 1$$

$$x = -2 \quad (1)$$

Section 4 continues on the following page.

2. Brian is currently 9 times as old as Alfred.

In 4 years, Brian will be 5 times as old as Alfred.

Calculate their current ages.

(Hint: Let A be the current Alfred's age and B be the current Brian's age.)

$$B = 9A \quad (1)$$

also

$$B + 4 = 5(A + 4)$$

Substitute (1) into this

$$9A + 4 = 5A + 20$$

$$\therefore 4A = 16$$

$$\therefore \boxed{A = 4} \quad (1)$$

$$\therefore B = 9 \times 4$$

$$\boxed{B = 36} \quad (2)$$

Section 4 continues on the following page.

3. The tickets for a music concert are \$ x for an adult and \$ y for a child. 3

The tickets of 3 adults and 17 children costs \$123 less than those of 12 adults and 5 children.

If the tickets of 6 adults and 9 children costs \$456, how much will it cost to buy the tickets of **23 adults** and **47 children** for the concert?

$$3x + 17y + 123 = 12x + 5y$$

$$\therefore 9x - 12y = 123$$

$$\therefore \boxed{3x - 4y = 41} \quad (1)$$

Also.

$$6x + 9y = 456$$

$$\therefore \boxed{2x + 3y = 152} \quad (2)$$

① mark for two equations

By solving (1) and (2) simultaneously.

$$\boxed{x = 43}$$

$$\boxed{y = 22}$$

① mark for solutions.

For 23 adults and 47 children,

$$\text{total cost} = 23 \times 43 + 47 \times 22$$

$$= \boxed{\$2023} \quad (1)$$

End of Section 4

